

Surface Form or Content? Decomposing Separability in AI-Generated Text Detection Benchmarks

Anonymous Author(s)

Affiliation withheld for double-blind review

Abstract—A detector that scores 99% on a machine-generated-text benchmark may have modelled how machine language differs from human language, or how one corpus happened to be punctuated and encoded, and the headline number does not distinguish the two. We propose a decomposition that measures the difference and apply it to two widely used corpora. A *surface-only* model reads 47 orthographic features and never sees a word, a *content-only* model reads text with punctuation, casing and non-ASCII characters stripped away, and a *full* model is a fine-tuned transformer on raw text. Across DAIGT V2 and HC3, five model families and eight configurations per corpus, the two benchmarks separate. On HC3 the surface and content arms are indistinguishable under a paired test, 3.20% against 3.26% error, so orthography alone matches an unfiltered bag-of-words over the whole vocabulary. On DAIGT V2 content is stronger by a factor of 7.9 in error. Both arms can read document length, a documented confound on both corpora, and closing that channel widens the DAIGT V2 gap to 10.6 times and turns HC3’s parity into a 2.91-point advantage for orthography. Repeating the decomposition over five group-aware partitions leaves the HC3 null significant on none of the five with the sign changing between them, which is what a null looks like and an underpowered real effect does not. A SHAP attribution over the surface arm locates the asymmetry, since one feature, the rate of spaces preceding punctuation, dominates HC3 at 2.3 times the next feature while no orthographic feature dominates DAIGT V2. Three controls change what the remaining results mean. BERT emits identical token identifiers for strings differing only by that whitespace cue, so it cannot represent the cue and still reaches 0.9916 weighted F1, making the cue sufficient in isolation but unnecessary in practice. A pipeline control confirms this null is not a cleaning step that failed to execute. A label-free control shows these models assign the human label to uniform random character strings in 100% of cases at 0.98 mean confidence, so an adversarial collapse to 0.3644 weighted F1 cannot be read as a targeted vulnerability. Surface form is informative on both corpora, and we claim only that it reaches parity with content on one of them.

Index Terms—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation, SHAP, paired significance testing

I. INTRODUCTION

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and frequently above 99% [1], [2], [3], [4]. Read directly, those numbers say the task is close to solved on the domains these corpora represent. Read carefully, they say something weaker, because a benchmark score measures the separability of a particular corpus and not the capability a reader infers from it.

The difference matters because separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language, or by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second have been documented, including a whitespace convention in HC3 [5], a length confound in the same corpus [6], punctuation inconsistencies requiring standardisation before comparison [7], and prompt-specific collection shortcuts [8]. Each of those is a report of one artefact. What is missing is a way to ask how much, in total, a given benchmark’s separability owes to surface form rather than content, and to place two benchmarks on the same axis so they can be compared.

This paper proposes that measurement and applies it. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model reads text after punctuation, casing and non-ASCII characters have been stripped away. A *full* model is a fine-tuned transformer on raw text. The gap between the first two bounds how much of a benchmark a detector could pass without reading language at all. Because the first two arms share a classifier family, they are directly comparable to each other, and the third is reported as a reference point rather than as a matched arm.

Applied to DAIGT V2 and HC3 across five model families, the answer differs sharply between the two corpora. On HC3 the surface-only and content-only arms are indistinguishable, 3.20% against 3.26% test error, and a paired test over 10,732 documents cannot separate them. On DAIGT V2 content is stronger by a factor of 7.9 in error, 0.99% against 7.86%. The same contrast appears in the model comparison. The best classical configuration on DAIGT V2 falls 0.0007 weighted F1 short of the best transformer, a gap a paired test on the two models’ own predictions cannot separate from zero, and falls short by an error-rate factor of 16 on HC3.

That DAIGT V2 tie is a statement about text budget rather than architecture, and we measure it rather than assume it. The classical model reads a whole essay while the transformer reads its first 128 tokens, which is roughly a third of the mean essay. Refitting the classical grid on the exact span each transformer’s own tokeniser kept removes the tie, and held to one text budget the transformer leads on both corpora.

We also report two negative results, because each corrects a

claim that an uncontrolled version of this analysis had reached and that the surrounding literature invites. The whitespace cue that dominates discussion of HC3 turns out to be sufficient in isolation yet unnecessary in practice, since BERT’s tokeniser provably cannot represent it and BERT still reaches 0.9916 weighted F1 on that corpus. And a character-level perturbation that drives DeBERTa from 0.9972 to 0.3644 weighted F1 looks like a targeted vulnerability but is not distinguishable from one, because the same model labels uniform random character strings as human in 100% of cases at 0.98 mean confidence. We report the methodological requirement rather than the vulnerability.

Our contributions are four. The first is the decomposition itself, stated formally in Section III-D as a reusable measurement over any human-versus-machine corpus, together with the length control that stops its two arms sharing a channel. The second is its application to two widely used benchmarks, with a complete eight-configuration model comparison, per-model ROC and area analysis, confusion structure, cross-corpus transfer over three seeds, and a per-feature SHAP attribution that locates which orthographic property carries the asymmetry. The third is a set of three controls, on tokenisation, on pipeline execution and on label-free inputs, each of which overturns a conclusion the uncontrolled version of this study had reached. The fourth is a paired statistical treatment, exact-binomial McNemar together with a paired bootstrap interval, applied uniformly so that no comparison in the paper rests on a difference in the fourth decimal place.

The rest of the paper is organised as follows. Section II places the work against prior detection results, artefact reports and robustness studies. Section III describes the corpora, the split, the models and the decomposition, and states the statistics. Section IV reports the model comparison, the decomposition and its controls, the transfer and ensemble results, and a consolidated view. Section V discusses what the measurement does and does not license, and Section VIII records where every number in the paper comes from.

II. RELATED WORK

Detection on these benchmarks. Guo et al. [1] introduce HC3 and report a RoBERTa detector at 99.82 F1, establishing both the corpus and the accuracy regime the field has worked in since. Su et al. [2] show that regime is narrower than it appears, since semantic-invariant tasks drop accuracy to roughly 91.7%. Zhou [3] reaches competitive DAIGT V2 accuracy with classical models over character n-grams and no transformer at all, a result consistent with ours but reported without the surface-content separation that would explain it, and without matching the text budget the transformers were given. Kubrusly et al. [9] likewise report strong bag-of-words results on HC3 without isolating what those features encode. Ansary [4] and Lamsiyah et al. [10] report near-ceiling DAIGT-family results, both in-distribution only. Annepaka et al. [11] mix linguistic features with a transformer for 99.45% on HC3, combining the two sources our decomposition separates. Yadagiri et al. [12] pair DeBERTa with auxiliary

stylistic features, which is the closest prior design to ours in its ingredients and the furthest in its purpose, since it combines the two channels to raise a score where we hold them apart to measure one. Socolof et al. [13] detect with style embeddings alone and argue for the interpretability that buys, a motivation we share, though a learned style embedding cannot be shown to exclude content the way a hand-built orthographic vector can. Broader surveys of the area [14], [15] catalogue detector families and threat models without asking what a given corpus is separable by, which is the question this paper takes up. Wang et al. [16] extend supervised detection to Chinese, a reminder that every result here is an English-language result and that the orthographic conventions our surface arm reads are not language-independent.

Competition solutions on DAIGT V2. The corpus was assembled for a public competition, and the solutions it produced are informative about what the corpus rewards. The winning entry [17] and several strong public ones [18], [19] lean on character n-grams, byte-pair tokenisation fitted to the test distribution and ensembles of gradient-boosted classical models rather than on larger language models. Independent replications reach the same place, with classical pipelines competitive against deep ones [20], [21], [22]. That pattern is usually read as evidence that classical models suffice for this task. Our decomposition suggests a second reading that is compatible with it and easier to act on, namely that a corpus whose surface arm alone reaches 0.9214 rewards models good at surface statistics, and that a leaderboard is therefore partly a measurement of the corpus rather than of the method.

Zero-shot and watermarking approaches. A parallel line of work detects without a trained classifier. GLTR [23] visualises per-token likelihood under a reference model, and DetectGPT [24] exploits the local curvature of the log-probability surface around machine-generated text. Watermarking [25] inserts a detectable signal at generation time instead of recovering one after the fact. These methods sidestep corpus-specific surface artefacts by construction, since they never fit to a training corpus, which is precisely why they are a useful contrast to the supervised setting we study. Our measurement does not apply to them directly, and it is not intended to.

Documented artefacts in these corpora. Tian et al. [5] identify the HC3 whitespace convention, observe that a detector could score well by exploiting it, and release a cleaning kit. Their contribution is the artefact and the remedy. They do not quantify how much of the corpus’s total separability surface form carries, and they do not test whether a given detector can represent the cue at all, which is the question our tokenisation control answers and which turns out to change the conclusion for BERT. Baidya et al. [6] document a length confound in HC3 and control it by length-matching, again one artefact treated in isolation, and it is the confound our own two arms turn out to share and that Section IV-F closes. Ardeshirifar [7] standardises punctuation and contractions before comparing handcrafted features to deep models, establishing the practice as necessary without measuring what it removes. Park et al. [8] show HC3-trained detectors rely on prompt-specific collection

shortcuts. Where these identify individual cues, we measure the aggregate, which is what a practitioner comparing two benchmarks needs.

Shortcut learning and dataset artefacts elsewhere. The pattern is not specific to text detection. Torralba and Efros [26] showed that vision datasets are identifiable from their own images, so a classifier can learn the collection rather than the category. In natural language inference, Gururangan et al. [27] and Poliak et al. [28] found that a model reading only the hypothesis reaches far above chance, and Niven and Kao [29] traced apparent argument comprehension to lexical cues that vanish under an adversarial split. Geirhos et al. [30] give the general account. Our surface-only arm is the same instrument as a hypothesis-only baseline, transferred to this task, and it plays the same role, which is to establish how far a model can get without the capability the benchmark is taken to measure. What differs is that a hypothesis-only baseline removes a span of text, while ours removes a property of it, so the two arms partition the signal along orthography rather than along position.

Robustness and its controls. Antoun et al. [31] attack HC3-trained detectors with misspellings and homoglyphs and report a fall from 99.88 F1 to 33.57% accuracy on a hand-built adversarial set. Huang et al. [32] report comparable degradation under systematic character perturbation. Neither reports a label-free control, and our results suggest such collapses cannot be separated from a generic degenerate response to unreadable input without one, since our own models label random character strings as human with near-maximal confidence. This is a methodological gap rather than a dispute about their measurements. Sadasivan et al. [33] argue on theoretical grounds that reliable detection becomes hard as generators approach human text, and Krishna et al. [34] show paraphrasing evades detectors in practice. Lai et al. [35] show ensembling preserves in-distribution accuracy while still degrading under shift, so the fragility is not single-model variance. Zhang et al. [36] reframe detection as a two-sample test, sidestepping single-document shortcuts at the cost of needing a population rather than a document. Borile and Abrate [37] ablate roughly twenty feed-forward neurons for up to 6.9 points of out-of-distribution gain, evidence that some in-distribution performance rests on localised corpus-specific features.

Fairness of the resulting decisions. Liang et al. [38] show that several deployed detectors misclassify writing by non-native English speakers as machine-generated at high rates, and that trivial rewriting removes the effect. That result bears directly on ours. A detector whose separability comes substantially from orthographic regularity is a detector whose errors will concentrate on writers whose orthographic habits differ from the corpus it was fitted to. We do not measure that effect here, and we note the connection because it is the most consequential reason to care which arm a benchmark’s score is coming from.

Cross-corpus evaluation. Alikhanov et al. [39] merge HC3 and DAIGT V2 into a single 124,195-sample corpus and

evaluate under a topic-held-out split, reporting 82.87% to 88.86% accuracy against the near-ceiling numbers usual in-distribution. Their design asks whether one model can serve both corpora. Ours asks what each corpus is separable by, which is why the two results sit alongside one another rather than replacing one another. Lu et al. [40] show detectors can be evaded by prompting alone, a threat model orthogonal to the corpus-property question studied here.

III. METHOD

A. Corpora

DAIGT V2 [41] contains 44,868 argumentative student essays, 27,371 human-written and 17,497 machine-generated by a mixture of 2023-era systems including GPT-3.5, Llama-2, Mistral-7B and Falcon-180B, assembled for a public detection competition [42]. The human side derives from the PERSUADE corpus of student argumentative writing [43]. HC3 [1] contains question-answer pairs from five English sources, contrasting human answers with a single generator, GPT-3.5-Turbo-0301, over 85,449 rows [44]. The two differ in almost every respect that matters here, which is why they are a useful pair. DAIGT V2 has many generators, one genre and long documents. HC3 has one generator, five source domains and short ones.

Both corpora were class-balanced by downsampling the majority class, giving 34,994 DAIGT V2 rows and 53,806 HC3 rows. Balancing is not cosmetic. It fixes the score of a degenerate all-one-class predictor at 0.333 weighted F1 rather than at something respectable, which is what makes the label-free control in Section IV-I readable. Training partitions are balanced to within 0.002 of parity.

TABLE I
CORPUS STATISTICS AFTER BALANCING. DUPLICATION IS MEASURED OVER THE UNBALANCED CORPUS BY EXACT MATCH ON WHITESPACE-NORMALISED, LOWERCASED TEXT. LEAKAGE IS THE SHARE OF TEST DOCUMENTS WHOSE EXACT TEXT ALSO APPEARS IN TRAINING OR VALIDATION, MEASURED ON THE BALANCED SAMPLE UNDER EACH SPLIT RULE. THE GROUP-AWARE RULE IS THE ONE EVERY NUMBER IN THIS PAPER USES.

	DAIGT V2	HC3
Rows, as published	44,868	85,449
Rows, balanced	34,994	53,806
Train / validation / test	25,196 2,800 6,998	38,785 4,289 10,732
Duplicate rows	6 (0.01%)	6,118 (7.16%)
Duplicate groups	6	5,986
Cross-label duplicate texts	0	0
Collection artefact rows	none found	483
Test leakage, group-aware split	0 (0.00%)	0 (0.00%)
Test leakage, naive stratified	0 (0.00%)	570 (5.30%)

A duplicate-detection pass over the unbalanced corpora finds DAIGT V2 essentially free of internal duplication, six rows in six groups, with no text appearing under both labels. HC3 carries 6,118 duplicate rows in 5,986 groups, plus 476 rows labelled machine-generated and 7 labelled human that are collection artefacts, typically stored API failure messages. Table I gives the figures.

B. Partitioning

That duplication is why the split is group-aware, and the choice is worth more than it sounds. Documents are grouped by an MD5 hash of their whitespace-normalised, lowercased text, and whole groups are assigned to a single partition, so no two documents with identical normalised content are placed on opposite sides of a boundary. The partition is 72/8/20 rather than 80/10/10, because a fifth is taken for test first and a tenth of the remaining four-fifths for validation.

Measured directly, the group-aware rule leaks no HC3 test document into training or validation, 0 of 10,732, whereas a plain stratified split of the same balanced sample leaks 570 of 10,762, which is 5.30%. DAIGT V2 leaks nothing either way. No number in this paper is inflated by memorised duplicates. The split is built once and reused unchanged by every model and every training seed, so a comparison between two models is never also a comparison between two evaluation sets.

C. Models and configurations

Five model families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each run under bag-of-words and TF-IDF for six classical configurations per dataset. Earlier work at full scale reported only the winning representation per model, so we report the complete grid. Two are transformers, `bert-base-uncased` [45] and `microsoft/deberta-v3-base` [46], fine-tuned with AdamW [47] over a grid of learning rate in $\{2e-5, 3e-5\}$, batch size in $\{16, 32\}$ and weight decay in $\{0.01, 0.1\}$, which is sixteen configurations per dataset, with the operating point selected on validation weighted F1 and early stopping at patience two.

DeBERTa-v3 is the variant chosen for three reasons that matter to this study rather than for its leaderboard position. Its disentangled attention separates content from position [48], which is the relevant inductive bias when the question is how much of a corpus is separable by form. Its ELECTRA-style pretraining [49] is more sample-efficient at a comparable parameter budget. And its SentencePiece tokeniser [50] encodes leading whitespace, whereas BERT’s WordPiece [51] does not, which makes the pair a controlled contrast on exactly the cue Section IV-H examines. Implementations are HuggingFace Transformers [52] and scikit-learn [53].

Classical models receive text through the normalisation pipeline described below, which lowercases, removes every character outside the basic Latin alphabet, drops English stopwords and lemmatises. Transformers receive raw text, whitespace-normalised only, truncated to 128 tokens. That asymmetry is deliberate, since stripping punctuation and casing removes signal a pretrained transformer can use, but it is also a confound for any classical-versus-transformer comparison, and Section IV-C measures it rather than assuming it away.

D. The surface-content decomposition

The central instrument of this paper separates how much of a benchmark’s human-versus-machine separability is available from surface form alone and how much from content alone. Fig. 1 shows the design.

Write x for a document and $y \in \{0, 1\}$ for its label, with 1 denoting machine-generated. Let $\mathcal{D}_{tr}$ and $\mathcal{D}_{te}$ be the fixed training and test partitions. Two maps are defined on the raw text. The surface map ϕ_S sends a document to $\mathbb{R}^{47}$, a vector of orthographic statistics described in Section III-F, and reads no word identity whatsoever. The content map ϕ_C first applies the normalisation

$$\nu(x) = \text{collapse}(\text{map}(\text{lower}(x), \sigma)), \quad (1)$$

where `lower` lowercases, `map` applies the character rule σ that keeps a character when it is a lowercase Latin letter or whitespace and replaces it with a space otherwise, and `collapse` reduces runs of whitespace to a single space. The normalised text is then sent to raw bag-of-words counts over the training vocabulary. Punctuation, casing and non-ASCII characters cannot survive ν , so they cannot enter the content arm. Word identity never enters ϕ_S .

Each arm fits one logistic regression on the same training partition,

$$h_a = \arg \min_h \sum_{(x,y) \in \mathcal{D}_{tr}} \ell(h(\phi_a(x)), y) + \frac{1}{2C} \|h\|_2^2, \quad (2)$$

for $a \in \{S, C\}$, with ℓ the logistic loss and C fixed at the library default for both arms. Sharing the classifier family and the regularisation strength is what makes the two arms comparable, since any difference between them is then a difference in what the representation carries rather than in how it is fitted.

The quantity we report is the error of each arm on the held-out partition,

$$\varepsilon_a = \frac{1}{|\mathcal{D}_{te}|} \sum_{(x,y) \in \mathcal{D}_{te}} \mathbf{1}[h_a(\phi_a(x)) \neq y], \quad (3)$$

and the separability gap between them,

$$\Delta = \varepsilon_S - \varepsilon_C. \quad (4)$$

A gap near zero says orthography alone reaches the same accuracy as content alone on that corpus. A large positive gap says content carries what orthography does not. The surface arm’s own error ε_S answers a second and more direct question, namely how much of the benchmark a detector could pass without reading language at all.

Two properties of Δ should be stated plainly. It is a comparison between two specific arms and not a partition of the total separability, so a richer surface featurisation would lower ε_S and a stronger content model would lower ε_C . Both arms therefore bound rather than decompose in the strict sense, and Δ should be read as a corpus-level comparison under a fixed pair of instruments. It is also not a claim that either corpus is clean, since ε_S is far below chance on both.

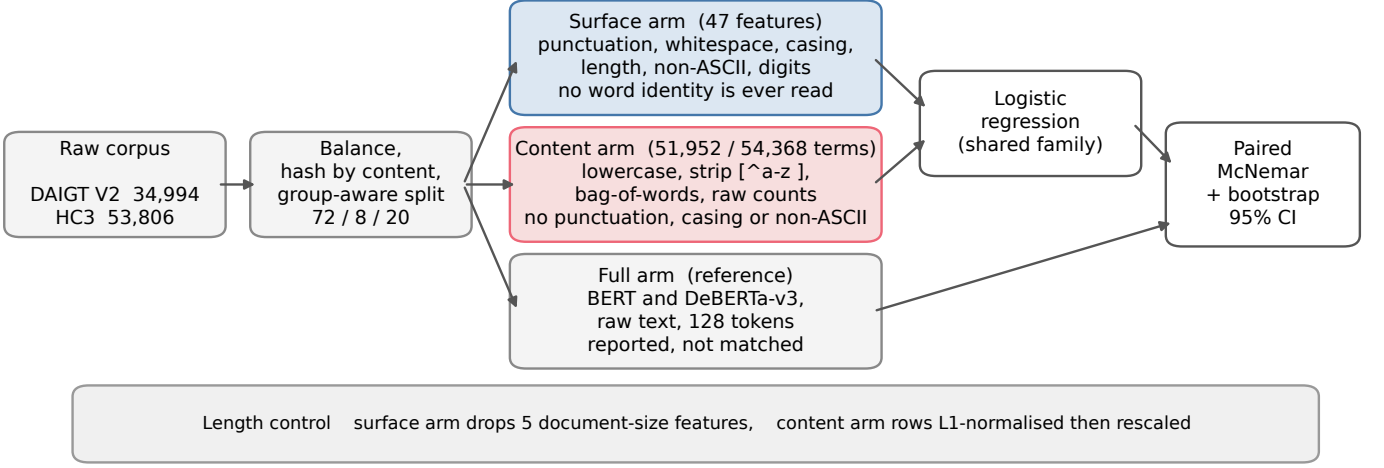


Fig. 1. The decomposition. One balanced, duplicate-group-aware partition feeds three arms. The surface arm reads 47 orthographic statistics and no word identity. The content arm reads an unfiltered bag-of-words after punctuation, casing and non-ASCII characters have been removed. The full arm is a fine-tuned transformer on raw text and is reported as a reference point, not as a matched arm. Surface and content share one classifier family so they are directly comparable, and every comparison between them is made with a paired test on the same documents. The length control at the bottom closes the one channel both primary arms can otherwise read. Counts on the diagram are read from the artefacts the experiment wrote.

The *full* arm is the fine-tuned transformer on raw text. It differs from the other two in model family, in parameter count and in the amount of text it reads, so it is reported as a reference point and we draw no conclusion from a transformer-to-classical difference alone without first matching the text budget.

E. The length channel and how it is closed

One channel survives into both arms as defined above. The surface arm reads character, word and sentence counts directly, and raw bag-of-words rows sum to document length, so ϕ_C can recover it too. Length is a documented confound on both corpora [6], so we close it rather than assume it away.

On the surface side we drop the five features whose magnitude scales with document size, leaving 42 rates and ratios, and write ϕ_S^{-L} for the reduced map. Every dropped feature has a rate twin in the same vector, so the cue itself is not removed along with the length. On the content side we normalise each row to unit total and then rescale by the mean training document length $\bar{n}$,

$$\phi_C^{-L}(x) = \bar{n} \cdot \frac{\phi_C(x)}{\|\phi_C(x)\|_1}. \quad (5)$$

The rescaling is not cosmetic. Rows that sum to one shrink every feature by roughly $\bar{n}$, so at a fixed C the effective penalty becomes far heavier and the arm collapses for reasons of scale rather than of length. We report both variants in Section IV-F and read only the rescaled one, and we treat the unrescaled arm as an instrument artefact rather than a result.

We also fit a *length-only* arm on the three pure document-size features, so the strength of the shared channel is measured directly instead of being inferred from what removing it costs the other two.

F. The surface feature set

Table II lists the families. The set is deliberately plain. Every feature is a count or a rate computable in one pass over

TABLE II
THE 47 SURFACE FEATURES, BY FAMILY. NONE READS WORD IDENTITY. THE FIVE FEATURES MARKED AS SIZE-SCALING ARE THE ONES THE LENGTH CONTROL REMOVES, AND EACH HAS A RATE TWIN THAT REMAINS. TWO FEATURES ARE CONSTANT ON THE DAIGT V2 TEST SPLIT AND ONE ON HC3, WHICH IS RECORDED BECAUSE A FEATURE WITH NO VARIANCE CONTRIBUTES NOTHING AND SHOULD NOT BE COUNTED AS EVIDENCE OF COVERAGE.

Family	Count	Notes
Space before punctuation	2	raw count is size-scaling
Document size	3	characters, words, sentences
Word shape	1	mean word length
Casing	3	uppercase, capitalised, all-caps
Non-ASCII and emoji	2	emoji count is size-scaling
Digits	1	rate per 1,000 characters
Whitespace layout	2	double spaces, newlines
Sentence structure	1	mean sentence length
Per-character punctuation rates	32	one per punctuation mark
Total	47	5 size-scaling, 42 rates and ratios

the raw string, with no model, no lexicon and no language identification, which is what makes the measurement cheap enough to run on any new corpus before committing to it.

Fig. 2 shows how much these features overlap. They are close to unredundant, which matters for two reasons. It means the surface arm’s performance is not one cue counted 47 times, and it means the length control removes a small and identifiable part of the vector rather than most of it in disguise.

G. Metrics

We report weighted F1 as the headline metric, because it is what the prior work on these corpora reports, alongside error rate because F1 differences compress at this accuracy ceiling and a change from 0.9916 to 0.9972 reads as small while the error rate it corresponds to falls by a factor of three. For class

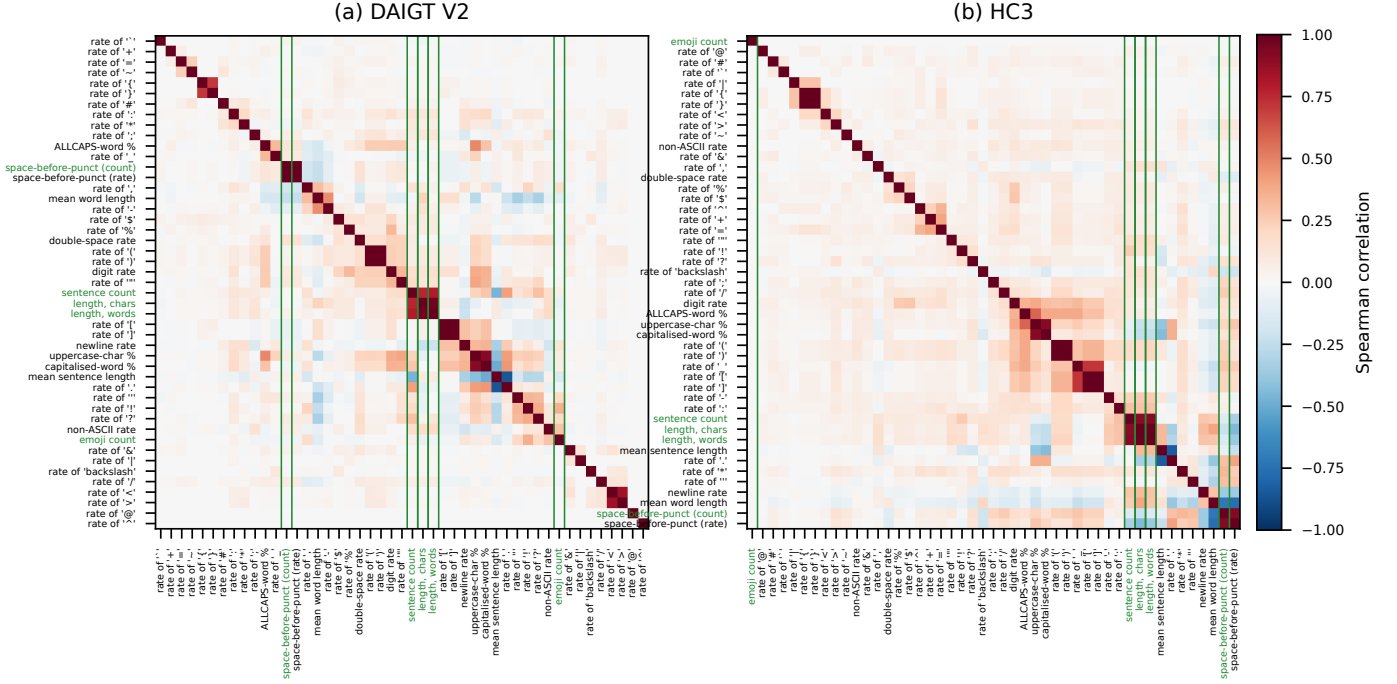


Fig. 2. Spearman correlation among the 47 surface features on each test split, with rows and columns ordered by average-linkage clustering on $1 - |\rho|$ so correlated blocks are adjacent. Features the length control removes are outlined and labelled in green. The set is close to uncorrelated on both corpora, with mean absolute off-diagonal correlation 0.062 on DAIGT V2 and 0.084 on HC3, and only 10 of 1,081 pairs on DAIGT V2 and 14 of 1,081 on HC3 exceeding 0.5 in absolute value. The visible blocks are the ones a reader would predict, namely the bracket and parenthesis rates moving together and the document-size features moving with the sentence count. Features with no variance on a test split have undefined correlation and are drawn as zero.

c with precision P_c and recall R_c , support n_c and $N = \sum_c n_c$,

$$F1_c = \frac{2P_c R_c}{P_c + R_c}, \quad F1_w = \sum_c \frac{n_c}{N} F1_c, \quad F1_m = \frac{1}{|C|} \sum_c F1_c. \quad (6)$$

We report both the weighted and the macro form throughout. On balanced test splits they coincide, and they do coincide to four decimal places in every row we report, which is the check that no result in this paper is produced by class skew. We also report the area under the ROC curve, because a threshold-free ranking measure can disagree with a thresholded one, and on DAIGT V2 it does.

H. Paired statistics

Every reported figure comes from a committed script that writes JSON, and every transformer baseline is read from the deployed checkpoint’s own record rather than from the maximum over its hyperparameter grid. An earlier version of this analysis compared a grid maximum against a single fixed configuration and reported differences that did not exist, which is why baselines are checkpoint-matched throughout.

Seed spread over three seeds is reported as a range and only for the cell it was measured on, never borrowed across datasets or model families, since fine-tuning outcomes vary materially with initialisation and data order [54], [55]. A range over three runs is an order-of-magnitude indication and not a test statistic, so no comparison in this paper rests on one. We do not run parametric tests at $n \leq 30$.

Where two models are compared on the same test set, the comparison is paired, because the two predictions are made on

the same documents and treating them as independent samples discards that structure [56]. Let b be the number of documents the first model classifies correctly and the second does not, and c the reverse. Under the null that the two have equal error, the discordant pairs split evenly, so

$$b \mid (b + c) \sim \text{Binomial}(b + c, \frac{1}{2}), \quad (7)$$

and we report the exact two-sided binomial p -value [57] rather than the chi-squared approximation, which is unreliable when $b + c$ is small, as it is for several comparisons here.

An exact test reports whether a difference is distinguishable from zero, not how large it is, so each comparison also carries a paired bootstrap interval on the error difference [58]. Writing $e_i^{(1)}$ and $e_i^{(2)}$ for the two models’ per-document error indicators, we resample document indices with replacement $B = 10,000$ times and take the empirical 2.5th and 97.5th percentiles of

$$\hat{\Delta}^{(b)} = \frac{1}{n} \sum_{i \in I_b} (e_i^{(1)} - e_i^{(2)}), \quad (8)$$

where I_b is the b th resample. Resampling index sets rather than each model’s errors independently is what preserves the pairing.

A bootstrap resamples one test set and therefore says nothing about how the corpus was partitioned in the first place. Because the central claim of this paper is a null on HC3, and a null is exactly the kind of claim a single lucky partition can manufacture, the whole decomposition is repeated over five group-aware partitions of the same balanced sample, varying only the split seed. Section IV-G reports what that shows.

I. Controls

Three controls guard the specific claims made below, and each was added after an uncontrolled version of the corresponding experiment produced a conclusion that did not survive.

The first is a *pipeline-fires* control. Our whitespace cleaning produces bit-identical predictions for BERT on HC3, which taken alone is indistinguishable from a cleaning step that silently failed to run. We therefore apply the same pipeline to DAIGT V2, where it also removes emoji that WordPiece does represent, and confirm that predictions there do change. The pipeline fires, and the null is specific to whitespace.

The second is a *tokenisation* control. Whether a detector can exploit a character-level cue is a property of its tokeniser and is directly testable, so we compare token identifier sequences for minimally different strings rather than reasoning about what the model probably does.

The third is a *label-free* control. We score 400 inputs per condition that carry no human-versus-machine information at all, including uniform random character strings, token-shuffled text, punctuation-only strings, a repeated single token and the empty string. Without this control a confident prediction on corrupted input is indistinguishable from a learned response to it, which is the distinction Section IV-I turns on.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 but not on HC3

TABLE III

ALL MODEL CONFIGURATIONS, BOTH DATASETS. TRANSFORMER ROWS ARE THE DEPLOYED CHECKPOINTS, NEVER A GRID MAXIMUM, AND CLASSICAL ROWS ARE REFITS ON THE SAME SPLIT. ERROR RATE IS GIVEN BESIDE F1 BECAUSE F1 COMPRESSES AT THIS CEILING. BOLD MARKS THE BEST CLASSICAL AND THE BEST OVERALL ENTRY PER DATASET COLUMN. MACRO F1 EQUALS WEIGHTED F1 TO FOUR DECIMALS IN EVERY ROW, SO NO RESULT IS PRODUCED BY CLASS SKEW.

Model	Rep.	DAIGT V2		HC3	
		F1	err %	F1	err %
Naive Bayes	BoW	0.9591	4.09	0.8713	12.87
Naive Bayes	TF-IDF	0.9577	4.23	0.8670	13.28
Logistic Reg.	BoW	0.9893	1.07	0.9551	4.49
Logistic Reg.	TF-IDF	0.9857	1.43	0.9365	6.35
SVM	BoW	0.9870	1.30	0.9475	5.25
SVM	TF-IDF	0.9910	0.90	0.9449	5.51
BERT	raw	0.9916	0.84	0.9916	0.84
DeBERTa	raw	0.9917	0.83	0.9972	0.28

Table III reports every configuration on both datasets. The two benchmarks differ in a way no single headline number reveals. On DAIGT V2 the best classical configuration, an SVM over TF-IDF at 0.9910 weighted F1, sits 0.0007 below the best transformer at 0.9917. On the same test documents the two disagree on 99 cases, 47 to 52, so the exact McNemar test returns $p = 0.69$ and the paired bootstrap puts the error difference at +0.07 percentage points with a 95% interval of $[-0.21, +0.34]$ that contains zero. On HC3 the same comparison is 0.9551 against 0.9972, with 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of

+4.21 points with interval $[+3.82, +4.61]$. The error-rate ratio there is sixteen to one.

Fig. 3 gives the corresponding ROC curves in the region that matters for a deployed detector, where false positives are expensive because they are accusations. The visual separation on HC3 and the absence of it on DAIGT V2 restate the table without the compression that F1 imposes near the ceiling.

Ranking by area does not reproduce ranking by F1. On DAIGT V2, BERT attains the higher area, 0.999641 against 0.999144, while DeBERTa attains the higher F1. Neither ordering is real, since the two models disagree on only 75 documents, 37 to 38, at $p = 1.00$. On HC3 the pair separates cleanly at $p < 10^{-6}$. We report both measures for this reason, and we would draw no conclusion from either ordering on DAIGT V2.

Fig. 4 shows the same curves unzoomed. The comparison between it and Fig. 3 is itself a small methodological observation, since at full scale the two corpora look alike and every model looks solved.

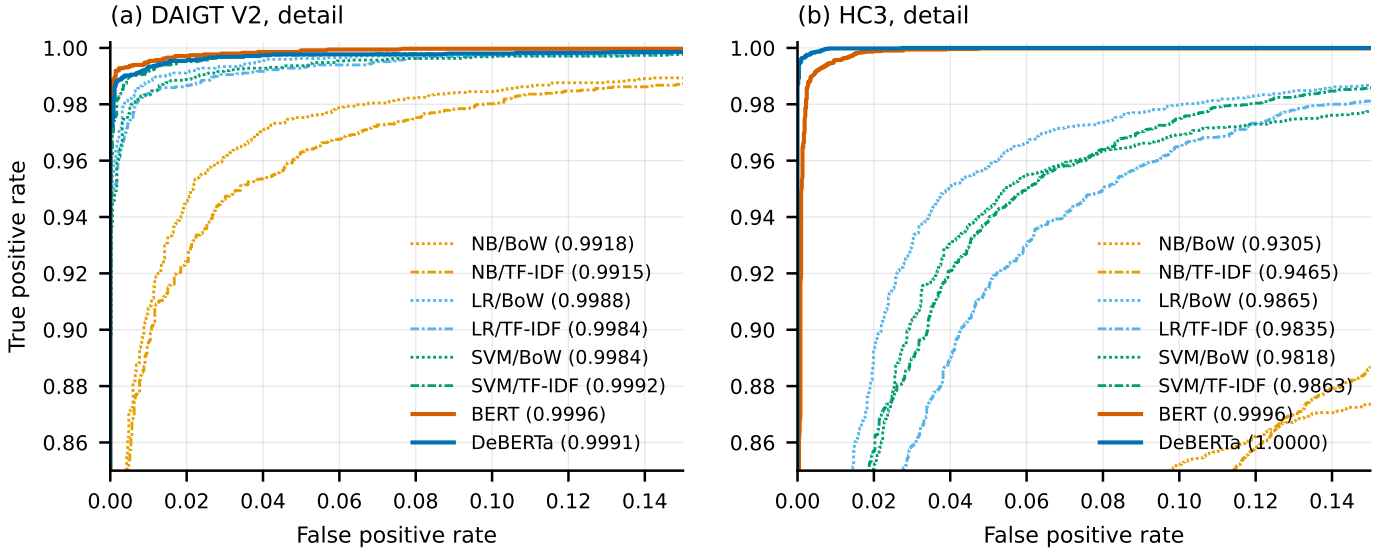


Fig. 3. ROC curves for all eight model configurations, restricted to the low-false-positive region where the models actually differ. Areas under the full curve are given in the legend. On DAIGT V2 in panel (a) every configuration converges toward the corner and the classical models are close behind the transformers. On HC3 in panel (b) the two transformers separate sharply from all six classical configurations, which is the same split Table III shows and the reason a single averaged benchmark score over both corpora would hide it. Colours are colourblind-safe and each series carries a distinct dash pattern, so the ordering survives greyscale reproduction.

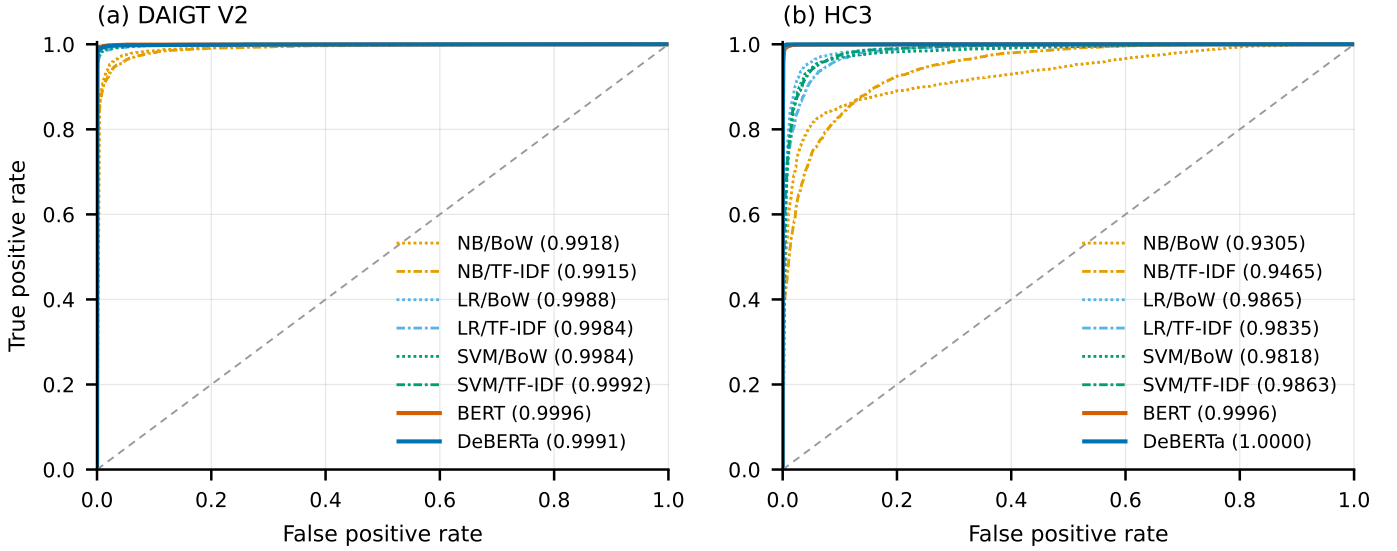


Fig. 4. The same eight configurations over the full ROC range. At this scale every curve is close to the corner on both corpora, which is what a headline accuracy figure reports and what motivates the low-false-positive view in Fig. 3. The figure is included because the compressed view is the one most often published, and the contrast between the two panels is the point.

B. Error structure is one-directional on neither corpus

Fig. 5 shows where the remaining errors fall. On unperturbed test data the four deployed checkpoints distribute their errors roughly evenly between calling a human document machine-generated and the reverse. That symmetry is the baseline against which the perturbation result in Section IV-I should be read, because the collapse reported there is not a uniform loss of accuracy but a wholesale movement of one class into the other.

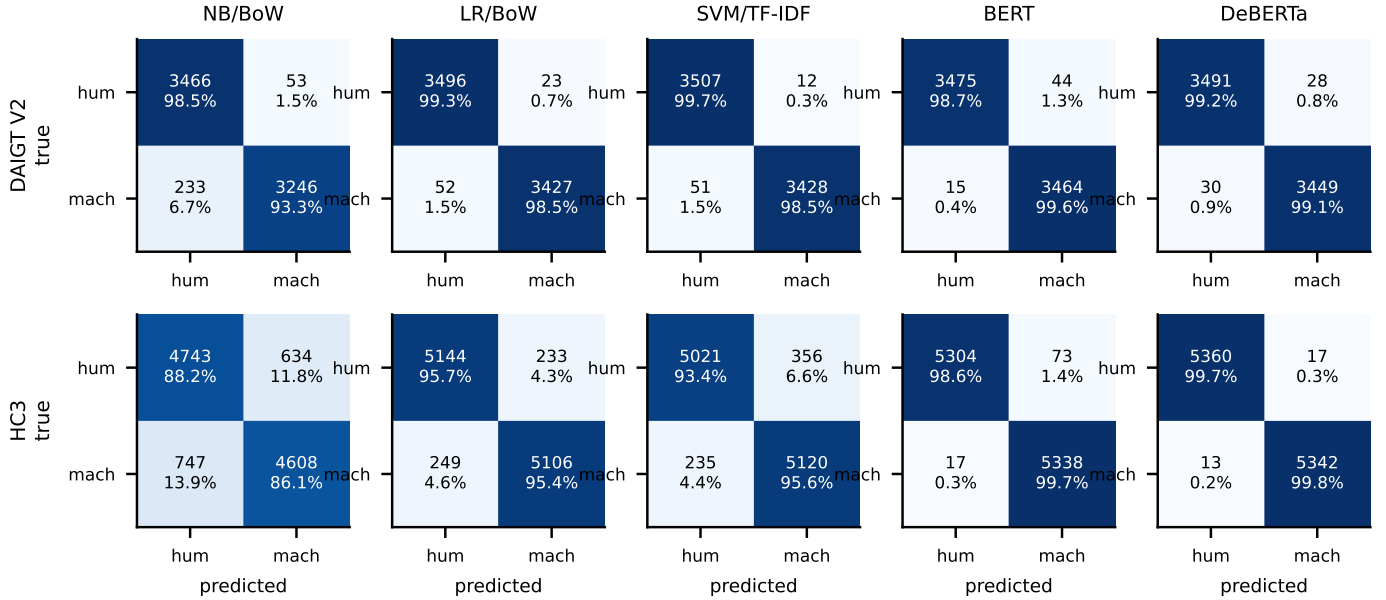


Fig. 5. Confusion matrices for the deployed BERT and DeBERTa checkpoints on both corpora, counts normalised within each true-label row. Errors are close to symmetric between the two directions on all four panels, so neither model is trading one error type for the other at its operating point. This matters for reading Section IV-I, where the same models under perturbation produce a sharply one-directional matrix instead.

C. The DAIGT V2 tie is a text budget, not an architecture result

TABLE IV

CLASSICAL MODELS REFITTED ON THE EXACT CHARACTER SPAN EACH TRANSFORMER’S OWN TOKENISER RETAINED, SO BOTH SIDES READ THE SAME TEXT. THE WINDOW IS DEFINED BY OFFSET MAPPING WITH NO DECODE ROUND-TRIP. HELD TO ONE BUDGET THE TRANSFORMER LEADS ON BOTH CORPORA, AND THE DAIGT V2 TIE IN TABLE III DISAPPEARS. THE TWO TOKENISERS DEFINE NEARLY THE SAME WINDOW AND RETURN THE SAME CONCLUSION.

Data	Window	chars		best classical err %		transf. err %
		kept	full text	128-tok		
DAIGT V2	BERT	0.335	0.90	2.10		0.84
DAIGT V2	DeBERTa	0.341	0.90	2.09		0.83
HC3	BERT	0.746	4.49	5.66		0.84
HC3	DeBERTa	0.756	4.49	5.72		0.28

That DAIGT V2 tie is not a matched comparison, and reading it as a statement about architecture would be wrong. The classical models receive the whole document while the transformers receive 128 tokens, which is 33.5% of a mean DAIGT V2 essay against 74.6% of a mean HC3 answer. Table IV reports what happens when the asymmetry is removed by refitting the classical grid on the exact span each checkpoint’s own tokeniser kept.

The best DAIGT V2 configuration then rises from 0.90% to 2.10% error against BERT’s 0.84%, a difference of 1.257 points with interval $[+0.900, +1.629]$ at $p < 10^{-6}$, and the DeBERTa window returns the same 1.257. On HC3 the gap only widens, from 4.49% to 5.66%. Held to one text budget the transformers lead on both corpora.

What the unmatched comparison shows is narrower and still worth stating, namely that a bag-of-words model reading a whole essay matches a transformer reading its opening. That is a claim about information access rather than about

architecture, and on a corpus of long documents it is arguably the more useful of the two.

D. On HC3, orthography alone matches content alone

TABLE V

SURFACE-CONTENT DECOMPOSITION. SURFACE-ONLY USES 47 ORTHOGRAPHIC FEATURES AND NEVER READS A WORD. CONTENT-ONLY IS UNFILTERED BAG-OF-WORDS AFTER PUNCTUATION, CASING AND NON-ASCII CHARACTERS ARE STRIPPED. THE LOWER BLOCK CLOSES THE DOCUMENT-LENGTH CHANNEL THAT BOTH PRIMARY ARMS CAN OTHERWISE READ. BOLD MARKS THE BETTER ARM IN EACH BLOCK AND DATASET COLUMN. THE TRANSFORMER IS A REFERENCE POINT, NOT A MATCHED ARM.

Arm	DAIGT V2		HC3	
	F1	err %	F1	err %
surface-only	0.9214	7.86	0.9680	3.20
content-only	0.9901	0.99	0.9674	3.26
full transformer	0.9917	0.83	0.9972	0.28
<i>length channel closed</i>				
length-only (3 feats)	0.7540	24.59	0.8475	15.23
surface-only, no length	0.9007	9.93	0.9662	3.37
content-only, length-norm.	0.9906	0.94	0.9371	6.28

Table V gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content. They disagree on 625 documents, split 316 to 309, so the exact McNemar test returns $p = 0.81$ and the paired bootstrap puts the difference at -0.07 percentage points with a 95% interval of $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary.

On DAIGT V2 the same two arms separate by a factor of 7.9 in error, 7.86% against 0.99%, and there the separation is unambiguous, with 569 discordant documents split 44 to 525, $p < 10^{-6}$, and a difference of $+6.87$ points with interval $[+6.23, +7.52]$.

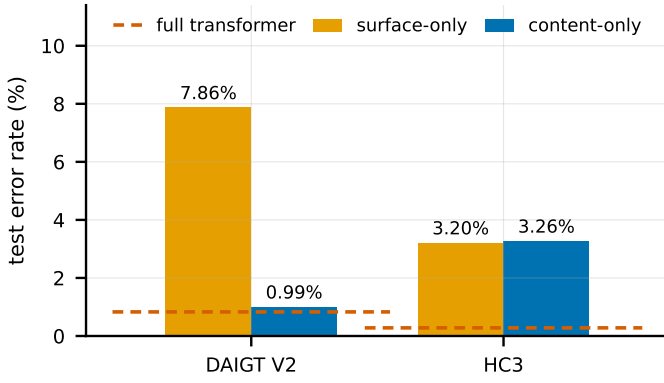


Fig. 6. The two primary arms and the transformer reference, per corpus. The HC3 pair is level and the DAIGT V2 pair is not, which is the paper’s central observation in one view. The transformer bar is drawn in grey to mark that it is a reference point rather than a matched arm, since it differs from the other two in model family, parameter count and the amount of text it reads.

Fig. 6 shows the same three arms per corpus. Surface form is informative on both benchmarks, since DAIGT V2’s surface-only arm reaches 0.9214 weighted F1, far above the 0.50 a coin would give and far above the 0.333 a degenerate single-class predictor would give. The finding is not that one corpus is clean and the other is not. It is that only on HC3 does surface rise to parity with content.

The content-only arm reaches 0.9901 on DAIGT V2 against the transformer’s 0.9917, within 0.16 points of error, under the same unmatched text budget as Section IV-C. On HC3 the transformer improves on either arm by roughly 0.03 weighted F1, which is consistent with it combining information neither arm holds alone.

E. One orthographic feature carries the HC3 result

TABLE VI

TOP FIVE SURFACE FEATURES BY MEAN ABSOLUTE SHAP VALUE, PER CORPUS, WITH THE SIGN OF THE FITTED COEFFICIENT. ON HC3 ONE FEATURE LEADS THE NEXT BY A FACTOR OF 2.3 AND THE TOP FOUR ALL PUSH TOWARD THE HUMAN CLASS. ON DAIGT V2 THE LEADING FEATURES ARE DOCUMENT SIZE AND WORD SHAPE AND THE LARGEST IS 2.8, A THIRD OF HC3’S LEADER, WITH NO COMPARABLE CONCENTRATION.

Data	Feature	mean SHAP	drives
DAIGT V2	length, characters	2.77	human
DAIGT V2	mean word length	1.88	machine
DAIGT V2	uppercase-char %	1.74	human
DAIGT V2	length, words	1.66	machine
DAIGT V2	capitalised-word %	1.49	machine
HC3	space-before-punct, rate	7.81	human
HC3	length, words	3.35	machine
HC3	length, characters	1.97	human
HC3	space-before-punct, count	1.20	human
HC3	rate of ‘)’	1.01	human

The decomposition in Section IV-D says how much orthography carries. It does not say which orthographic property carries it, and the two corpora turn out to differ there as sharply as they differ in the aggregate. Table VI lists the leading features by per-feature Shapley attribution [59] over the surface arm, and Fig. 7 shows the full distributions. Because the arm is a linear model, the attributions are computed exactly rather than

sampled, so the ranking below carries no estimation error of its own.

On HC3 the single largest contributor is the rate of spaces preceding punctuation, at mean absolute SHAP 7.81 against 3.35 for the next feature, and its fitted coefficient is negative, so its presence drives the human label. That is the collection convention Tian et al. [5] identified, recovered here without being looked for and quantified against the rest of the feature set. Section IV-H then shows why its presence does not license the conclusion a reader would reach next.

On DAIGT V2 the leading contributors are document length, mean word length and casing statistics, none of which exceeds 2.8 and none of which is a collection convention in the same sense. The surface arm on that corpus is reading a diffuse stylistic signal rather than a single artefact, which is consistent with its much weaker showing against the content arm.

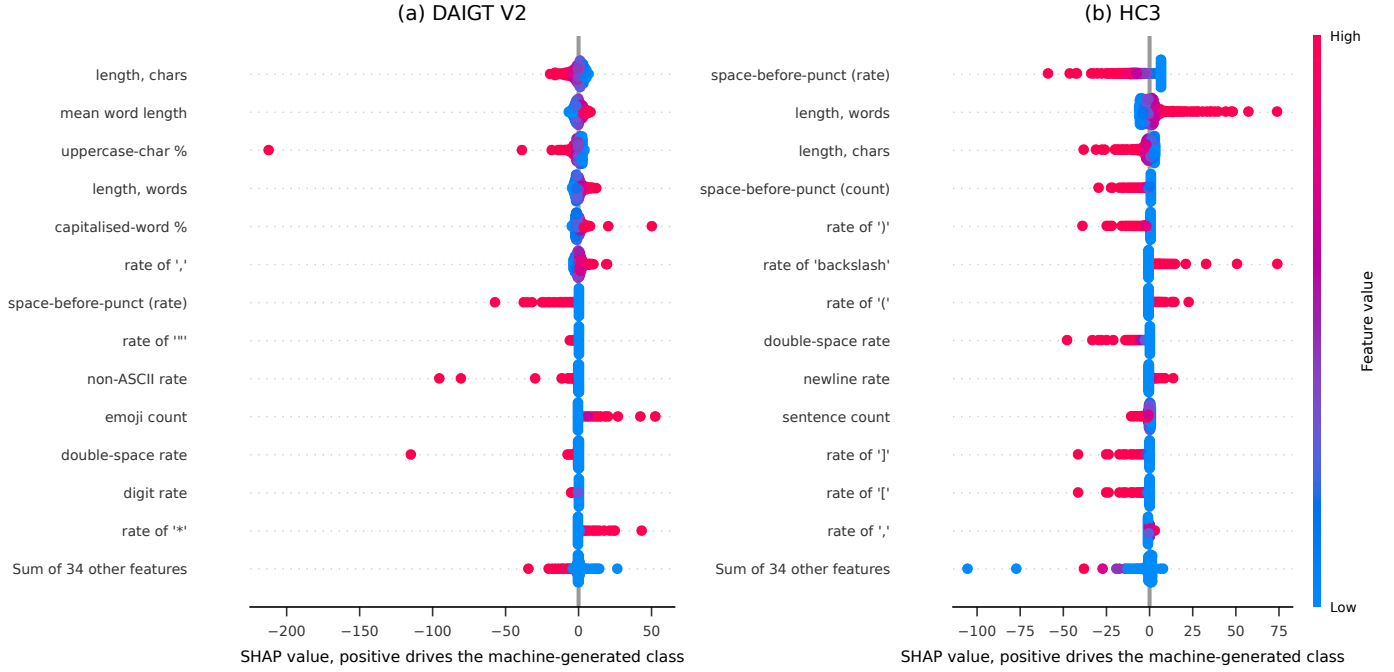


Fig. 7. SHAP attributions over the surface-only arm, computed exactly with a linear explainer on 2,000 held-out documents per corpus. Each point is one document, positioned by its contribution to the machine-generated class and coloured by the feature’s value. The two corpora look nothing alike. On HC3 in panel (b) the rate of spaces preceding punctuation dominates every other feature and pushes hard toward the human class. On DAIGT V2 in panel (a) no orthographic feature dominates, and the leading contributors are document length and word shape rather than a single collection convention.

F. Neither result is an artefact of document length

Both arms can read document length, so the comparison above could be a length comparison in disguise. Length alone reaches 24.59% error on DAIGT V2 and 15.23% on HC3, well short of either arm but far from nothing, which is why the channel is closed rather than argued away.

Closing it on both sides sharpens rather than weakens the two conclusions. On DAIGT V2 the length-free content arm loses nothing, 0.94% against 0.99% at $p = 0.71$, while the length-free surface arm rises to 9.93%, widening content’s advantage from 7.9 to 10.6 times. On HC3 the surface arm barely moves, from 3.20% to 3.37%, while the content arm rises to 6.28%, so orthography no longer merely matches content but beats it by 2.91 points, with interval $[+2.37, +3.44]$ at $p < 10^{-6}$. Removing length costs the content arm eighteen times what it costs the surface arm, which is the opposite of what a length-driven surface result would produce.

One caution on the instrument belongs here, because it nearly produced a third false conclusion. Normalising bag-of-words rows to unit total without restoring their scale shrinks every feature by roughly the mean document length, so at a fixed regularisation strength the penalty becomes far heavier and the arm collapses to 9.69% on DAIGT V2 and 14.25% on HC3. Read uncritically that looks like a dramatic length effect. It is an artefact of scale, visible in the optimiser trace, since the unrescaled arm converges in 16 to 24 iterations against 121 to 192 for the raw-count arm. Every figure quoted above uses rows rescaled to the raw-count arm’s magnitude, and the unrescaled arm is reported only as a negative control.

G. The HC3 null survives repartitioning

TABLE VII
THE DECOMPOSITION REPEATED OVER FIVE GROUP-AWARE PARTITIONS OF THE SAME BALANCED SAMPLE, VARYING ONLY THE SPLIT SEED. SPLIT VARIANCE IS BETWEEN 0.17 AND 1.29 PERCENTAGE POINTS, SO THE SINGLE-SPLIT FIGURES IN TABLE V WERE NOT MISLEADING, BUT THE PARITY CLAIM RESTS ON THIS TABLE RATHER THAN ON ONE PARTITION.

Arm	DAIGT V2	HC3
	mean err % (range)	mean err % (range)
surface-only	7.15 (6.74–7.86)	3.16 (3.07–3.33)
content-only	0.82 (0.66–0.99)	3.24 (3.06–3.41)
surface-only, no length	9.76 (9.47–9.93)	3.26 (3.14–3.37)
content-only, length-norm.	0.84 (0.77–0.94)	6.02 (5.82–6.28)
length-only	24.31 (23.56–24.85)	15.32 (15.23–15.49)

A null on one partition is the kind of claim a single lucky split can manufacture, so the decomposition is repeated over five group-aware partitions of the same balanced sample. Table VII reports the result. Seed 42 reproduces the single-split numbers exactly, which is the harness self-check that makes the other four readable.

Content leads on DAIGT V2 on all five partitions, by 6.00 to 6.87 points, at $p < 10^{-6}$ throughout. On HC3 the surface-content difference is significant on none of the five, with p equal to 0.81, 0.27, 0.34, 0.23 and 0.78, and the difference changes sign between partitions, ranging from -0.30 to $+0.27$ points. That sign instability is the strong form of the null, because an underpowered real difference keeps its sign while a genuine null does not. The length-free comparison, by contrast, is significant on all five and keeps its sign, from -2.67 to -2.91 points, so the advantage orthography gains once length is removed is not a partition effect either.

H. A model that cannot represent the cue reaches 0.9916 anyway

The most-discussed HC3 artefact is a space preceding punctuation in human answers [5], and Section IV-E confirms it is the single strongest orthographic feature on that corpus. In our test split it appears 10.745 times per human document against 0.013 per machine one, and in 88.7% of human documents against 0.3% of machine ones. DAIGT V2 carries the same cue far more weakly, 0.548 against 0.004 per document, in 17.1% against 0.2% of documents.

The natural next inference is that HC3-trained detectors exploit it. Tokenisation makes that inference testable rather than rhetorical, and the test does not support it.

BERT’s WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace. For the strings "the answer is simple ." and "the answer is simple." it emits identical identifier sequences, in three of three pairs tested, while DeBERTa’s SentencePiece encodes the leading whitespace and distinguishes all three. BERT cannot represent the cue at all and still reaches 0.9916 weighted F1 on HC3. The cue is therefore sufficient in isolation, since the surface arm that reads it reaches 0.9680, and unnecessary in practice, since a model blind to it reaches 0.9916.

This is also why removing the cue changes nothing for BERT. Applying whitespace cleaning to HC3 yields predictions bit-identical to the uncleaned run, verified by SHA-256 over the saved probability arrays. Taken alone that would be indistinguishable from a cleaning step that never executed. The same pipeline applied to DAIGT V2, where it also removes emoji that WordPiece does represent, does change BERT’s predictions. The pipeline fires, and the null is specific to whitespace.

We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue. The two models differ in tokeniser, architecture, pretraining corpus and parameter count, and this experiment separates none of those. What it establishes is narrower and, for anyone planning a cleaning ablation, more useful, namely that a null result from removing a cue is uninterpretable without checking whether the model could read the cue in the first place.

I. The adversarial collapse is not distinguishable from a degenerate response

TABLE VIII

LABEL-FREE CONTROL. PERCENTAGE OF INPUTS ASSIGNED THE *human* LABEL, WITH MEAN MAXIMUM SOFTMAX PROBABILITY IN PARENTHESES, ON 400 INPUTS PER CONDITION THAT CARRY NO HUMAN-VERSUS-MACHINE INFORMATION AT ALL. A DETECTOR WITH A CALIBRATED RESPONSE TO UNREADABLE INPUT WOULD NOT ANSWER THESE CONFIDENTLY IN EITHER DIRECTION. ALL FOUR CHECKPOINTS DO.

Checkpoint	rand. chars	shuffled	punct.
DAIGT V2 BERT	0% (0.91)	94.8% (0.99)	97.8% (0.71)
DAIGT V2 DeBERTa	100% (0.99)	100% (1.00)	100% (1.00)
HC3 BERT	51.0% (0.71)	98.5% (1.00)	100% (1.00)
HC3 DeBERTa	100% (0.98)	100% (1.00)	100% (0.99)

Perturbing HC3 test text with character-level typos at 10% of alphabetic characters drives DeBERTa from 0.9972 to 0.3644 weighted F1. The confusion matrix is one-directional, [[5376, 1], [5207, 148]], with human documents still classified almost perfectly and machine documents reassigned to human. Compared against the symmetric error structure in Fig. 5, that looks like a targeted vulnerability. It is not what the control shows.

Table VIII reports the label-free control. On inputs carrying no human-versus-machine information, the same model assigns the human label to uniform random character strings in 100% of cases at mean confidence 0.98, and to token-shuffled and punctuation-only text likewise. All four checkpoints label the latter two conditions human between 94.8% and 100% of the time. The behaviour is specific to neither HC3 nor the attack.

Three candidate mechanisms were tested and none survives. Majority-prior collapse is excluded by the training balance, which is 0.4981 against 0.5019 on DAIGT V2 and 0.4999 against 0.5001 on HC3. Cue injection is excluded because the perturbation does not create the whitespace artefact, moving the machine-side count only from 0.000 to 0.005. A learned association from disfluency to human authorship is unsupported, since subword fertility runs the wrong way, at 1.1192 for HC3 human text against 1.1980 for machine.

We therefore decline to read our own perturbation numbers as robustness measurements, and we report them in Table IX for completeness rather than as evidence. The experiment establishes a requirement on future work rather than a property of these benchmarks.

TABLE IX

PERTURBATION RESULTS FOR THE DEPLOYED CHECKPOINTS, REPORTED FOR COMPLETENESS. IN-DOMAIN FIGURES ARE EACH CHECKPOINT’S OWN SCORE, NEVER A GRID MAXIMUM. WE DO NOT INTERPRET THESE AS ROBUSTNESS MEASUREMENTS, FOR THE REASON TABLE VIII GIVES. BACK-TRANSLATION, WHICH LEAVES TEXT READABLE, DEGRADES ALL FOUR MODELS ONLY SLIGHTLY, AND IT IS THE CHARACTER-LEVEL ATTACKS THAT PRODUCE THE LARGE FALLS.

Data	Model	typo 5%	typo 10%	back-tr.
DAIGT V2	BERT	0.9106	0.7966	0.9856
DAIGT V2	DeBERTa	0.9836	0.9590	0.9767
HC3	BERT	0.4390	0.3746	0.9872
HC3	DeBERTa	0.5524	0.3644	0.9773

The contrast within Table IX is itself informative. Back-translation rewrites the text while leaving it readable and costs all four models between 0.004 and 0.020 weighted F1. Character-level corruption, which makes the text unreadable, is what produces the collapses. That pattern is consistent with the label-free reading and not with a targeted cue-removal reading.

J. Neither corpus transfers to the other

Table X reports transfer in both directions over three seeds. Every cell loses a large amount, between 0.0833 and 0.2019 mean weighted F1, which is the expected consequence if a substantial part of each corpus’s separability is corpus-specific. DeBERTa’s gap is roughly half BERT’s in both directions at the mean level. We report that as a mean-level pattern that

TABLE X

CROSS-CORPUS TRANSFER, THREE SEEDS PER CELL. EACH MODEL IS TRAINED ON ONE CORPUS AND EVALUATED ON THE OTHER WITH NO ADAPTATION. RANGES ARE THE OBSERVED MINIMUM AND MAXIMUM OVER THE THREE SEEDS AND ARE NOT CONFIDENCE INTERVALS.

Trained	Model	in-dom.	cross-dom. (range)	gap
DAIGT	BERT	0.9921	0.7902 (0.770–0.810)	0.2019
DAIGT	DeBERTa	0.9930	0.9096 (0.891–0.926)	0.0833
HC3	BERT	0.9930	0.8311 (0.813–0.852)	0.1619
HC3	DeBERTa	0.9970	0.8512 (0.830–0.887)	0.1458

survives three-seed averaging rather than as a confirmed effect, because with three seeds per cell the observed ranges for BERT’s two directions nearly touch, 0.182 to 0.222 against 0.140 to 0.182, and this project does not run parametric tests at that sample size.

K. The ensemble does not reliably improve on its stronger member

TABLE XI

SOFT-VOTE ENSEMBLE OF THE TWO DEPLOYED CHECKPOINTS, WITH THE MIXING WEIGHT CHOSEN ON VALIDATION ONLY AND APPLIED ONCE TO TEST. THE DAIGT V2 MIX SCORES HIGHER THAN EITHER MEMBER BUT ITS PAIRED INTERVAL CONTAINS ZERO. ON HC3 VALIDATION SELECTION ASSIGNS ZERO WEIGHT TO BERT, SO THE ENSEMBLE IS DeBERTa AND REPORTS DeBERTa’S SCORE.

Data	w_B	BERT	DeBERTa	ens.	95% CI (pp)
DAIGT V2	0.50	0.9916	0.9917	0.9936	[−0.39, +0.01]
HC3	0.00	0.9916	0.9972	0.9972	[0.00, 0.00]

Table XI reports a soft-vote ensemble over the two deployed checkpoints, with the mixing weight selected on validation weighted F1 and applied once to test. On DAIGT V2 an equal mix reaches 0.9936 against DeBERTa’s 0.9917, which reads as an improvement until the comparison is made on the documents themselves. The two disagree in a way that gives an exact McNemar p of 0.085 and a bootstrap interval on the error difference of [−0.39, +0.01] percentage points, which contains zero.

On HC3 the validation sweep places zero weight on BERT. The ensemble is DeBERTa, and its reported score is DeBERTa’s score. We record the degenerate weight rather than reporting the number as an ensemble result, because a soft-vote ensemble whose selected weight has discarded one member is not an ensemble.

Reporting either row as an improvement would be reading a difference in the fourth decimal place as a finding. We treat the ensemble as a discussion point, consistent with Lai et al. [35], who find that ensembling preserves in-distribution accuracy without addressing the shift that actually degrades these models.

L. A consolidated view

Fig. 8 collects the results above. Panel (c) is the one worth reading closely, because it is where the paper’s central claim is either supported or refuted. The DAIGT V2 series sits between six and seven points above zero on every partition. The HC3 series straddles zero and changes sign. Panel (f) shows the

same distinction in interval form, where the grey intervals are those that contain zero, and the two of them are exactly the two comparisons this paper declines to call differences.

V. DISCUSSION

The decomposition separates two benchmarks that a headline accuracy figure does not. On HC3, orthography alone matches content alone, 3.20% against 3.26% test error, and a paired test over 10,732 documents cannot distinguish that 0.07-point difference from zero, nor can the same test on any of five independent partitions. A fine-tuned DeBERTa improves on either arm by roughly 0.03 weighted F1. On DAIGT V2, content is stronger than orthography by a factor of 7.9 in error. The best classical model there is within 0.0007 weighted F1 of the best transformer, a gap the same test also cannot separate from zero, though only because the two models read different amounts of text. A practitioner choosing an evaluation corpus, or a reviewer reading a reported score, cannot recover that distinction from the score itself.

The attribution in Section IV-E says where the difference comes from. HC3’s surface arm is close to a single-feature model, since the rate of spaces before punctuation leads the next feature by a factor of 2.3 and the corpus carries that cue in 88.7% of human documents against 0.3% of machine ones. DAIGT V2’s surface arm reads a diffuse mixture of length, word shape and casing with no comparable leader. Those are different kinds of corpus even though both are separable by form, and the practical consequence differs accordingly. A single collection convention can be cleaned. A diffuse stylistic signal cannot, and it may not even be an artefact.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2’s surface-only arm reaches 0.9214 weighted F1, far above chance, so surface form is substantially informative on both. The finding is comparative, that surface reaches parity with content on one corpus and not the other, and it is bounded by the arms we built. A richer surface featurisation would raise both surface numbers and a stronger content model would raise both content numbers, so the arms bound rather than partition the separability, in the sense made explicit in Section III-D.

The two arms are also not disjoint as built, since document length reaches both. Closing that channel weakens neither conclusion. DAIGT V2’s content advantage widens from 7.9 to 10.6 times, and HC3’s parity becomes a 2.91-point advantage for orthography, because removing length costs the content arm eighteen times what it costs the surface arm. The control nearly produced a third false conclusion of its own, since normalising bag-of-words rows without restoring their scale collapses the arm through regularisation and reads as a length effect.

Two results here began as the opposite conclusion, and both are worth stating for what they say about method rather than about these corpora. A cleaning experiment that reports no change is uninterpretable without evidence that the pipeline ran at all, and a tokeniser that cannot represent the cue being cleaned makes the null a fact about the model rather than about

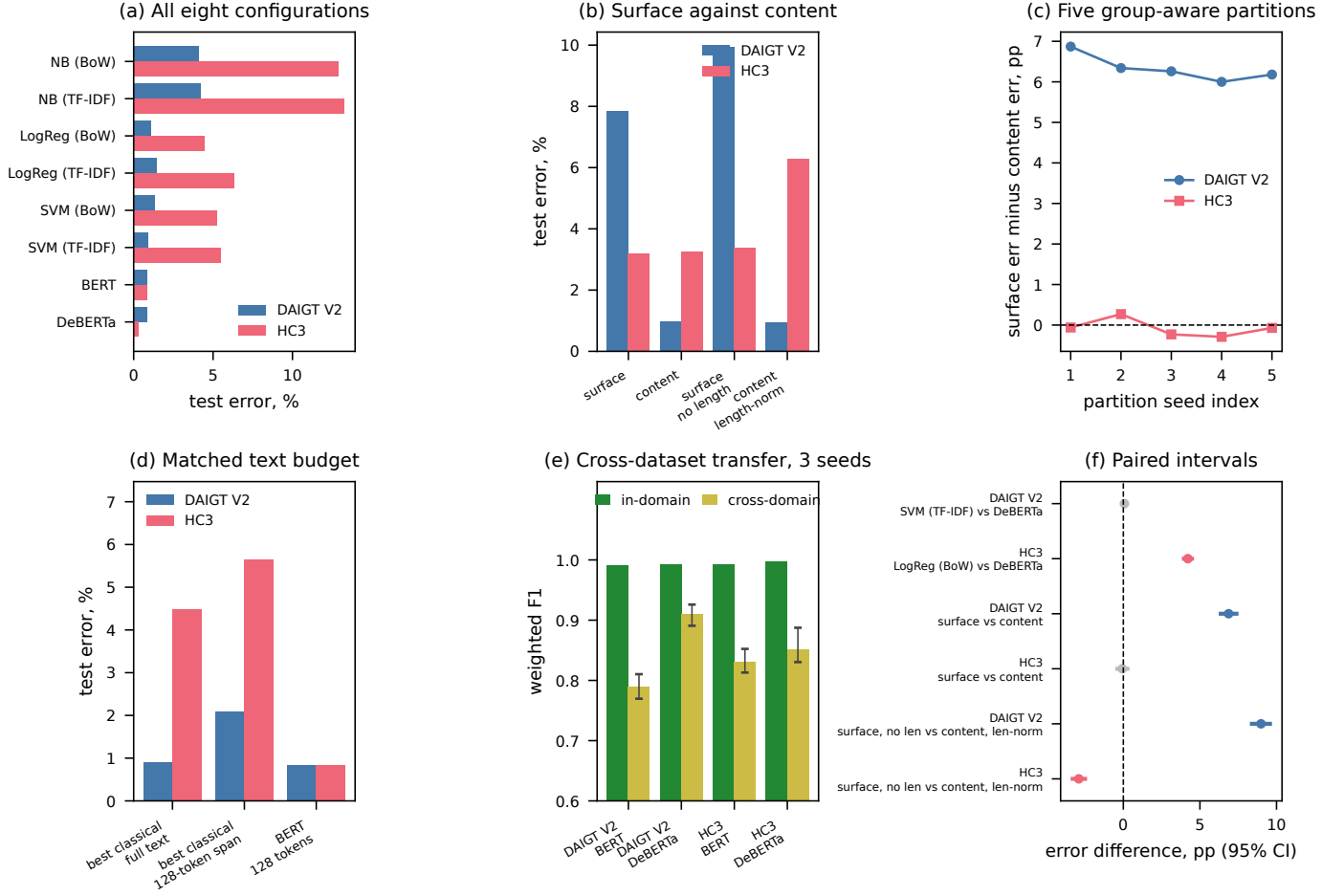


Fig. 8. Every headline result in one view. Panel (a) ranks all eight configurations by test error on both corpora. Panel (b) gives the decomposition with and without the length channel. Panel (c) plots the surface-minus-content error difference across five group-aware partitions, where the DAIGT V2 series sits far above zero on all five and the HC3 series crosses it. Panel (d) shows the matched text budget from Table IV. Panel (e) shows cross-corpus transfer with the observed three-seed range as error bars. Panel (f) shows the paired bootstrap intervals behind the significance claims, with intervals containing zero drawn in grey. Every value is read from the JSON the audit scripts wrote and none is entered by hand.

the corpus. A collapse under perturbation is uninterpretable without a label-free control, since a model that calls random strings human at 0.98 confidence will produce the same curve with no vulnerability behind it. We report both requirements rather than the findings they dissolve, and we suggest comparable published collapses be read with the second control in mind.

There is a connection to fairness that we did not measure and that seems to us the most consequential reason to care which arm a score comes from. Liang et al. [38] show deployed detectors misclassify non-native English writing as machine-generated at high rates. A detector whose separability rests substantially on orthographic regularity is a detector whose false positives should be expected to concentrate on writers whose orthographic habits differ from the corpus it was fitted to. Our measurement predicts which corpora carry that risk most sharply. It does not demonstrate the effect, and establishing it would need annotated author metadata that neither corpus provides.

VI. LIMITATIONS

Several limits bound every number above, and we state them rather than leaving them to be inferred.

The transformers see at most 128 tokens, so the DAIGT V2 transformer results describe classification from an essay’s opening rather than from the essay. We measure that rather than assume it, by refitting the classical grid on the same window, and it reverses the classical-versus-transformer reading on that corpus. The cross-corpus comparison in Table X remains not like-for-like in the amount of text each model receives.

Seed spread is estimated from three seeds, and one such estimate moved by 31% when its runs were repeated, because GPU training here is not bitwise deterministic and best-epoch selection can choose a materially different checkpoint on a rerun. No claim in this paper rests on a seed range, and every comparison uses a paired test on saved predictions instead.

The content arm keeps stopwords and inflections, so it is not a reduced content model. It is still a bag-of-words one, and word order and syntax are outside it, so it bounds content-carried separability from below for that reason rather than

through its preprocessing. A syntactic or contextual content arm would raise ε_C 's competitor and could narrow the DAIGT V2 gap.

The surface set is 47 hand-built features, two of which have no variance on the DAIGT V2 test split and one of which has none on HC3. It is deliberately cheap and it is certainly not exhaustive, so ε_S is an upper bound on surface-arm error rather than a measurement of everything surface form could carry.

HC3 has one generator collected at one time, and both corpora are English. We evaluate two corpora and five model families, and nothing here establishes that the pattern holds beyond them. In particular we do not claim that HC3 is representative of short-form question answering, or DAIGT V2 of student writing, only that these two widely used benchmarks differ from one another on the axis we measure.

VII. CONCLUSION

We proposed a decomposition that measures how much of a machine-generated-text benchmark's separability is available from surface form alone rather than from content, and applied it to DAIGT V2 and HC3 across five model families and eight configurations per corpus. On HC3 the two arms are indistinguishable under paired testing on five independent partitions, and a per-feature attribution shows a single collection convention accounts for most of the surface arm. On DAIGT V2 content leads by a factor of 7.9 in error, and the surface signal is diffuse rather than concentrated. Closing the document-length channel that both arms share strengthens both conclusions. Three controls, on tokenisation, on pipeline execution and on label-free inputs, each overturned a conclusion that this study had reached without them, and we report the controls as requirements rather than the findings they dissolved.

The measurement itself is the portable part. It needs no new annotation, it fits two logistic regressions, and it runs in minutes on a laptop. It yields one number per corpus that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language. We would encourage reporting it alongside any new detection benchmark, in the same way a hypothesis-only baseline is now reported alongside a natural language inference dataset [28].

VIII. REPRODUCIBILITY

Every number in this paper is produced by a committed script that writes a JSON file, and no number was transcribed from a session log. The mapping from claim to artefact is as follows, and the naming below is the naming in the released repository.

The corpus statistics and leakage figures in Table I come from the two full-corpus audits and from the split verifier, which measures leakage directly on both split variants rather than estimating it. The eight-configuration comparison in Table III and the ROC curves in Fig. 3 and Fig. 4 come from a single evaluation script that reads transformer scores from each deployed checkpoint's own record and refits the classical grid on the same partition. The decomposition in Table V and

Fig. 6, together with the length-controlled arms, comes from one script that also writes per-document predictions, which is what makes the paired tests in Section IV-D possible rather than asserted. The five-partition repeat in Table VII varies only the split seed and reproduces the seed-42 figures exactly, which is its self-check. The matched-budget comparison in Table IV uses offset mapping to recover the exact character span each tokeniser kept, with no decode round-trip. The attributions in Fig. 7 and Table VI use an exact linear explainer over the same surface featuriser the decomposition uses, imported rather than reimplemented so the two cannot drift apart. The label-free control in Table VIII and the perturbation results in Table IX come from two further scripts, and the ensemble in Table XI is computed from the deployed checkpoints' saved probability arrays with the mixing weight selected on validation only.

The partition is built once, by a script that asserts no duplicate-content group crosses a boundary, and is reused unchanged by every model and every training seed. Transformer training uses a fixed seed for initialisation and data order, with best-epoch selection on validation weighted F1 and early stopping at patience two. We record that GPU training in this environment is not bitwise deterministic, so a rerun of the same seed can select a different best epoch, and this is why no claim in the paper rests on a seed range.

Three practices are worth naming because each was adopted after an error that reached a draft. Transformer baselines are read from the deployed checkpoint's own record rather than from the maximum over its hyperparameter grid, after an ablation that compared a grid maximum against a fixed configuration reported differences that did not exist. Seed spread is quoted only for the cell it was measured on, after a spread measured on one corpus and model family was applied as a noise floor to a comparison on another. And a claimed leakage rate that had no located source file was removed and then measured, at which point it turned out to be 5.30% under a naive split and zero under the split this paper uses.

A single notebook reproduces the preprocessing, both transformers at their deployed configurations and the ensemble, reading each configuration from the checkpoint record at runtime rather than from a value typed into the notebook. Both corpora are public. Code is released for reproduction. The link is withheld for blind review.

REFERENCES

- [1] B. Guo, X. Zhang, Z. Wang, M. Jiang, J. Nie, Y. Ding, J. Yue, and Y. Wu, "How close is chatgpt to human experts? comparison corpus, evaluation, and detection," arXiv, 2023, preprint, arXiv:2301.07597.
- [2] Z. Su, X. Wu, W. Zhou, G. Ma, and S. Hu, "Hc3 plus: A semantic-invariant human chatgpt comparison corpus," arXiv, 2023, preprint, arXiv:2309.02731.
- [3] C. Zhou, "Exploiting machine learning model ensemble for AI-generated texts detection," *Transactions on Computer Science and Intelligent Systems Research*, vol. 5, 2024, aIDML 2024. [Online]. Available: <https://wepub.org/index.php/TCSISR/article/view/2382>
- [4] M. S. Ansary, "A multirepresentation stacked ensemble for AI-generated text detection," *International Journal of Intelligent Systems*, vol. 2026, no. 1, 2026.
- [5] Y. Tian, H. Chen, X. Wang, Z. Bai, Q. Zhang, R. Li, C. Xu, and Y. Wang, "Multiscale positive-unlabeled detection of AI-generated texts," arXiv, 2023, preprint, arXiv:2305.18149.

- [6] M. S. Baidya, S. S. Baidya, and C. Chawla, "Detecting the machine: A comprehensive benchmark of AI-generated text detectors across architectures, domains, and adversarial conditions," arXiv, 2026, preprint, arXiv:2603.17522.
- [7] R. Ardeshtirifard, "Comparing hand-crafted and deep learning approaches for detecting AI-generated text: performance, generalization, and linguistic insights," *AI and Ethics*, vol. 5, no. 4, pp. 4197–4209, 2025.
- [8] C. Park, H. J. Kim, J. Kim, Y. Kim, T. Kim, H. Cho, H. Jo, S. goo Lee, and K. M. Yoo, "Investigating the influence of prompt-specific shortcuts in AI generated text detection," arXiv, 2024, preprint, arXiv:2406.16275.
- [9] J. Kubrusly, E. D. dos Santos, and L. S. Pelegrino, "Detecting AI-generated texts using machine learning models," *Communications in Statistics: Case Studies, Data Analysis and Applications*, vol. 11, no. 4, pp. 495–512, 2025.
- [10] S. Lamsiyah, S. Ezzini, A. E. Mahdaouy, H. Alami, A. Benlahbib, S. E. Amrany, S. Chafik, and H. Hammouchi, "M-daigt: A shared task on multi-domain detection of AI-generated text," arXiv, 2025, preprint, arXiv:2511.11340.
- [11] Y. Annepaka, P. Kumar, Y. Poddar, P. Pakray, and C. Chunka, "Synergizing linguistic features and transformer networks for detecting AI-generated text," *Knowledge and Information Systems*, vol. 68, no. 1, 2026.
- [12] A. Yadagiri, S. Sai Teja, P. Pakray, and C. Chunka, "AI-generated text detection using DeBERTa with auxiliary stylistic features," in *Proceedings of the RANLP 2025 Workshop on Multi-Domain Detection of AI-Generated Text (M-DAIGT)*, 2025. [Online]. Available: <https://aclanthology.org/2025.ranlp-mdaiht.2/>
- [13] G. Z. Socolof and R. Kacholia, "Fast, interpretable AI-generated text detection using style embeddings," Stanford CS224N final project, 2024, not peer-reviewed. [Online]. Available: <https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1244/final-projects/GiuliaZoeSocolofRitikaKacholia.pdf>
- [14] G. Jawahar, M. Abdul-Mageed, and L. V. S. Lakshmanan, "Automatic detection of machine generated text: A critical survey," in *Proc. Int. Conf. Computational Linguistics (COLING)*, 2020, pp. 2296–2309.
- [15] E. Crothers, N. Japkowicz, and H. L. Viktor, "Machine-generated text: A comprehensive survey of threat models and detection methods," *IEEE Access*, vol. 11, pp. 70 977–71 002, 2023.
- [16] R. Wang, H. Chen, R. Zhou, H. Ma, Y. Duan, Y. Kang, S. Yang, B. Fan, and T. Tan, "Llm-detector: Improving AI-generated chinese text detection with open-source llm instruction tuning," arXiv, 2024, preprint, arXiv:2402.01158.
- [17] R. Biswas *et al.*, "Kaggle 'LLM – detect AI generated text', 1st place solution," Kaggle competition write-up, 2024, not peer-reviewed. [Online]. Available: <https://github.com/rbiswasfc/llm-detect-ai>
- [18] Zeyadusf, "DAIGT – catch the AI," GitHub, n.d., graduation project, not peer-reviewed; undated in source. [Online]. Available: <https://github.com/zeyadusf/DAIGT-Catch-the-AI>
- [19] A. Zain, S. Farooqui, and M. Rafi, "A multi-strategy approach for AI-generated text detection," arXiv, 2025, preprint, arXiv:2509.00623.
- [20] B. Neo, "Did AI write this? using classical machine learning," bitgrit Data Science Publication (Medium), n.d., industry blog post, not peer-reviewed; undated in source. [Online]. Available: <https://medium.com/bitgrit-data-science-publication/did-ai-write-this-7963a8d7ed2d>
- [21] "Supervised machine generated text detection using LLM," Worcester Polytechnic Institute MS thesis repository, 2024, mS thesis, not peer-reviewed; author name not identifiable from accessible metadata (PDF download link, no author field extractable) – do not fabricate, cite by title/institution only. [Online]. Available: <https://digital.wpi.edu/downloads/hh63t0231>
- [22] S. Arun, S. Purohit, and J. Pareek, "Analyzing AI-generated text using machine learning, deep learning and large language models," in *Advances in Smart Computing and Applications*, ser. Communications in Computer and Information Science. Springer Nature Switzerland, 2026, pp. 173–187.
- [23] S. Gehrmann, H. Strobelt, and A. M. Rush, "GLTR: Statistical detection and visualization of generated text," in *Proc. ACL: System Demonstrations*, 2019, pp. 111–116.
- [24] E. Mitchell, Y. Lee, A. Khazatsky, C. D. Manning, and C. Finn, "DetectGPT: Zero-shot machine-generated text detection using probability curvature," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023, pp. 24 950–24 962.
- [25] J. Kirchenbauer, J. Geiping, Y. Wen, J. Katz, I. Miers, and T. Goldstein, "A watermark for large language models," in *Proc. Int. Conf. Machine Learning (ICML)*, 2023, pp. 17 061–17 084.
- [26] A. Torralba and A. A. Efros, "Unbiased look at dataset bias," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2011, pp. 1521–1528.
- [27] S. Gururangan, S. Swayamdipta, O. Levy, R. Schwartz, S. R. Bowman, and N. A. Smith, "Annotation artifacts in natural language inference data," in *Proc. NAACL-HLT*, 2018, pp. 107–112.
- [28] A. Poliak, J. Naradowsky, A. Haldar, R. Rudinger, and B. Van Durme, "Hypothesis only baselines in natural language inference," in *Proc. Joint Conf. on Lexical and Computational Semantics (*SEM)*, 2018, pp. 180–191.
- [29] T. Niven and H.-Y. Kao, "Probing neural network comprehension of natural language arguments," in *Proc. ACL*, 2019, pp. 4658–4664.
- [30] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.
- [31] W. Antoun, V. Moulleron, B. Sagot, and D. Seddah, "Towards a robust detection of language model generated text: Is chatgpt that easy to detect?" arXiv, 2023, preprint, arXiv:2306.05871.
- [32] G. Huang, Y. Zhang, Z. Li, Y. You, M. Wang, and Z. Yang, "Are AI-generated text detectors robust to adversarial perturbations?" arXiv, 2024, preprint, arXiv:2406.01179.
- [33] V. S. Sadasivan, A. Kumar, S. Balasubramanian, W. Wang, and S. Feizi, "Can ai-generated text be reliably detected?" *arXiv preprint arXiv:2303.11156*, 2023.
- [34] K. Krishna, Y. Song, M. Karpinska, J. Wieting, and M. Iyyer, "Paraphrasing evades detectors of ai-generated text, but retrieval is an effective defense," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [35] Z. Lai, X. Zhang, and S. Chen, "Adaptive ensembles of fine-tuned transformers for llm-generated text detection," arXiv, 2024, preprint, arXiv:2403.13335.
- [36] S. Zhang, Y. Song, J. Yang, Y. Li, B. Han, and M. Tan, "Detecting machine-generated texts by multi-population aware optimization for maximum mean discrepancy," arXiv, 2024, preprint, arXiv:2402.16041.
- [37] C. Borile and C. Abrate, "How to generalize the detection of AI-generated text: Confounding neurons," in *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025, pp. 25 461–25 476. [Online]. Available: <https://aclanthology.org/2025.findings-emnlp.1388.pdf>
- [38] W. Liang, M. Yuksekgonul, Y. Mao, E. Wu, and J. Zou, "GPT detectors are biased against non-native English writers," *Patterns*, vol. 4, no. 7, p. 100779, 2023.
- [39] A. Alikhanov, A. Amangeldi, D. Demeubay, D. Akhmetzhan, N. Moldakhmetov, O. Polat, and G. Zharas, "AI generated text detection," arXiv, 2026, preprint, arXiv:2601.03812.
- [40] N. Lu, S. Liu, R. He, Q. Wang, Y.-S. Ong, and K. Tang, "Large language models can be guided to evade AI-generated text detection," arXiv, 2023, preprint, arXiv:2305.10847.
- [41] thedrcat, "DAIGT v2 train dataset," Kaggle, 2023. [Online]. Available: <https://www.kaggle.com/datasets/thedrcat/daigt-v2-train-dataset>
- [42] "LLM – detect AI generated text," Kaggle competition, 2023. [Online]. Available: <https://www.kaggle.com/competitions/llm-detect-ai-generated-text>
- [43] "PERSUADE 2.0 corpus," GitHub, 2023. [Online]. Available: https://github.com/scrosseye/persuade_corpus_2.0
- [44] "HC3 – human ChatGPT comparison corpus," Hugging Face Datasets, 2023. [Online]. Available: <https://huggingface.co/datasets/Hello-SimpleAI/HC3>
- [45] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [46] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2023, arXiv:2111.09543.
- [47] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2019, arXiv:1711.05101.

- [48] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-enhanced BERT with disentangled attention,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021, arXiv:2006.03654.
- [49] K. Clark, M.-T. Luong, Q. V. Le, and C. D. Manning, “ELECTRA: Pre-training text encoders as discriminators rather than generators,” in *Proc. Int. Conf. Learning Representations (ICLR)*, 2020, arXiv:2003.10555.
- [50] T. Kudo and J. Richardson, “SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing,” in *Proc. EMNLP: System Demonstrations*, 2018, pp. 66–71.
- [51] Y. Wu, M. Schuster, Z. Chen, Q. V. Le, M. Norouzi *et al.*, “Google’s neural machine translation system: Bridging the gap between human and machine translation,” *arXiv preprint arXiv:1609.08144*, 2016.
- [52] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue *et al.*, “Transformers: State-of-the-art natural language processing,” in *Proc. EMNLP: System Demonstrations*, 2020, pp. 38–45.
- [53] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [54] J. Dodge, G. Ilharco, R. Schwartz, A. Farhadi, H. Hajishirzi, and N. Smith, “Fine-tuning pretrained language models: Weight initializations, data orders, and early stopping,” arXiv, 2020, preprint, arXiv:2002.06305.
- [55] M. Mosbach, M. Andriushchenko, and D. Klakow, “On the stability of fine-tuning bert: Misconceptions, explanations, and strong baselines,” arXiv, 2020, preprint, arXiv:2006.04884.
- [56] T. G. Dietterich, “Approximate statistical tests for comparing supervised classification learning algorithms,” *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998.
- [57] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.
- [58] B. Efron, “Bootstrap methods: Another look at the jackknife,” *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.
- [59] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4765–4774.